Library Management System

Software Requirements Specification

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Prepared by:	Eshal P, Systems Analyst
Organization:	Mar Athanasius College of Engineering / XB Software Ltd.
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Revision History

Name	Date	Reason For Changes	Version	Date of Approval
Eshal P	May 15, 2026	Initial draft compilation and content integration.	0.1	Draft
Eshal P	May 17, 2026	Final structural verification, functional alignment, and template execution.	1.0	May 17, 2026

Table of Contents

1. Introduction

1.1 Purpose

The main objective of this document is to illustrate the requirements of the project Library Management system. The document gives the detailed description of both functional and non-functional requirements proposed by the client. The document is developed after a number of consultations with the client and considering the complete requirement specifications of the given Project. The final product of the team will be meeting the requirements of this document. This specification applies strictly to Version 1.0 of the web-based system deployment.

1.2 Document Conventions

For tracking the changes mark them in the document with color (changes, additions) or cross them (deleting). Standards headings are formatted sequentially, and bolding is explicitly utilized to denote user interface paths, entity definitions, and critical error-handling metrics.

1.3 Intended Audience and Reading Suggestions

This document is structured specifically for various operational and analytical stakeholders. It is recommended that overview sections (Sections 1 and 2) be read first by all parties before moving into specialized technical sections.

ID	Stakeholder	Description
S-1	Customer	Checking correspondence of business goals and functionality requirements to the expectations from implementing the product.
S-2	Development team	Forming the accurate vision of the project, detailed functional and nonfunctional requirements.
S-3	QA team	Making test-plans and test-cases.
S-4	PM/BA	Estimating the quote of the project, planning resources and the timeline of work.
S-5	User	Basing on this document the Terms of Service and the Privacy Policy are created.

1.4 Product Scope

The product is designed for both the Users and Library Admin. It will be a helpful product in a very effective way as it will reduce the tiresome workload from both Users and Library Admin. The system automates routine processes such as item registration, physical distribution tracking, fine accruals, and user database configurations, directly advancing corporate directives to modernize administrative infrastructure and optimize personnel allocations.

1.5 References

1. *IEEE Std 830-1998*, IEEE Recommended Practice for Software Requirements Specifications.
2. *Library Architecture Design & User Experience Guidelines*, Version 2.1, March 2025.
3. *MySQL Database Enterprise Security Standards Documentation*, Oracle Corp, 2024.

2. Overall Description

2.1 Product Perspective

The proposed Library Management System is an on-line Library Management System. This System will provide a search functionality to facilitate the search of resources. This search will be based on various categories viz. book name or the ISBN. Also Advanced Search feature is provided in order to search various categories simultaneously. Further the library staff personnel can add/update/remove the resources and the resource users from the system.

2.2 Product Functions

The system fundamentally partitions behaviors across core functional vectors centered around the administration tier (Librarian) and the consuming tier (Users). Below is the consolidated operational architecture map:

LIBRARIAN Functions:

- A librarian can issue a book to the student
- Can view The different categories of books available in the Library
- Can view the List of books available in each category
- Can take the book returned from students
- Add books and their information of the books to the database

• Edit the information of the existing books.

• Can check the report of the issued Books.

• Can access all the accounts of the students.

USERS Functions:

• Can view the different categories of books available in the Library

• Can view the List of books available in each category

• Can own an account in the library

• Can view the books issued to him

• Can put a request for a new book

• Can view the history of books issued to him previously

• Can search for a particular book

2.3 User Classes and Characteristics

ID	User classes	Description
U-1	Administrator / Librarian	A signed up user who has completed the account activation. He owns expanded rights inside the portal. He performs the content pre moderation, stock adjustments, and account verification.
U-2	Signed up user (Student)	A user of the portal who has completed the sign up on the portal and the account activation. Can hold checkouts and issue requests.
U-3	Not signed up user	A user of the portal who has completed neither the sign up on the portal, nor the account activation. He owns limited rights inside the portal, restricted entirely to public catalog search features.

2.4 Operating Environment

This system will be operated on any computer with the following minimum specifications:-

1. Windows XP Service Pack-3 OR higher versions of windows.
2. Computer hardware should be build on INTEL chipset.
3. Minimum free RAM of 128 MB.
4. Internet connectivity required.

2.5 Design and Implementation Constraints

The design & implementation constraints are:-

1. The system database used should be an open-source technology.
2. The system should be implemented in Java Technology
3. The downtime of the system should be less than 10 min.
4. RAM usage should not exceed 1024MB
5. This system software size should not exceed 1GB.

2.6 User Documentation

The user manual and help will be available online and can be accessed any time by any user. The manual will be updated on a regular basis so as to keep the contents up to date. This includes contextual help icons built natively within the administrator control panels.

2.7 Assumptions and Dependencies

It is assumed that the optimum internet connectivity speed will be more than 512Kbps. If the bandwidth is less than this then the transaction completion will take more time to processed and to be complete. Project execution relies fundamentally on the underlying open-source Java Virtual Machine remaining natively compatible with existing database drivers.

3. External Interface Requirements

3.1 User Interfaces

Various GUI elements like forms, images and standard buttons will be included in the User Interface. Screen components must map out responsive behaviors across modern enterprise desktop web applications, prioritizing low structural density and consistent input-form navigation pathways.

3.2 Hardware Interfaces

The system interacts over common networking architectures with dedicated hardware servers. It enforces physical compatibility layers via standard TCP/IP networking protocols to transmit transactional operations securely.

3.3 Software Interfaces

Software will work on Windows OS. The Database used will be an open-source database like MySql. And the system will run on Java Virtual Machine.

3.4 Communications Interfaces

This system will require web browser, internet connection which supports HTTP and server. It maps out secure layer integrations using transport controls over explicit endpoints, maintaining low-latency synchronization constraints across records.

4. Domain Model

The structural layout of the application consists of distinct logical entity mappings. Core entities include **Book** (attributes: BookName, ISBN, Category, AvailabilityStatus), **UserAccount** (attributes: UserID, AccountStatus, MaxAllowedBooks), and **TransactionRecord** (attributes: TransactionID, IssueDate, ReturnDate, FineAccrued).

5. System Features (Use Cases)

5.1 Use Case: Issue Book

Brief Description:	Allows the Librarian to register a book checkout session to an active student account database file.
Business Trigger:	The Student physically presents a book selection to the checkout counter area.
Preconditions:	The Librarian is successfully authenticated into the console, the user account is active, and the book availability status is true.

Line	System Actor Action (Librarian)	System Response
1	Initiates a book checkout request within the active console.	Prompts for Student ID and Book identification details.
2	Inputs Student ID and Book ISBN parameter.	Validates constraints, logs the relationship, and updates inventory records.
3	Confirms checkout finalization.	Displays verification receipt and updates return deadlines.
Post Condition:	The book instance status changes to unavailable, and a transaction log is successfully committed.	

5.2 Use Case: Return Book

Brief Description:	Processes a book return session, clearing user balance metrics and assessing fine metrics.
Preconditions:	System is up, and a valid checkout row maps to the incoming physical book object.

5.3 Use Case: Search Book Stock

Brief Description:	Enables both signed up and anonymous user layers to search categories synchronously.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

1. Login/Registration will not take more than 10 seconds.
2. Any financial transactions will not take more than 15 seconds.

6.2 Safety Requirements

Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. The backend processing layers must run memory isolation barriers to avoid memory overflow exceptions or thread starvation issues during processing iterations.

6.3 Security Requirements

1. Database will be secured by authentication process.
2. Unauthorized access will be avoided and will be tracked.
3. Database backup will be maintained.

6.4 Software Quality Attributes

1. System will be reliable.
2. System can be maintained easily.

7. Other Requirements

Internationalization parameters must permit Unicode UTF-8 field processing across all structural database tables. System logs must index transactional actions sequentially to meet institutional accountability frameworks.

8. Appendices

8.1 Appendix A: Glossary

Term	Description
SRS	Software Requirements Specification document structure.
ISBN	International Standard Book Number utilized for physical document cataloging.
JVM	Java Virtual Machine runtime platform context.

8.2 Appendix B: Analysis Models

Optional dynamic diagrams detailing state changes and class hierarchies will follow in engineering blueprints.

8.3 Appendix C: Interface Examples

Prototypical UI components follow standard text-field validation models across input constraints:

Field	Description	Constraints / Validation
Book Name / Task	Editable text entry form.	Max 100 characters; alphanumeric parameters only.
Status	System-rendered display column.	Non-editable; automated lookup response tracking.

8.4 Appendix D: To Be Determined List

All open architectural queries requiring structural client alignment are recorded here for system expansion passes.